

EXPERIMENT-11

IP Spoofing Using GitHub Tool in Kali Linux

Step 1: Start Environment

Open Oracle VirtualBox and start Kali Linux.

Step 2: Prepare Tool

Download or clone an IP spoofing tool from GitHub and install required dependencies.

Step 3: Configure Parameters

Set the target IP address, spoofed source IP, and network interface in the tool.

Step 4: Execute Tool

Run the tool to send packets with a forged source IP address.

Step 5: Monitor Traffic

Observe the network traffic using monitoring tools to verify spoofed packets.

Output

Packets are transmitted with modified (spoofed) source IP addresses and reflected in network traffic.

```
saira-banu@saira-banu-VirtualBox:~$ ip a
lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default
qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP gr
p default qlen 1000
    link/ether 08:00:27:53:06:f3 brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute enp0s3
        valid_lft 86245sec preferred_lft 86245sec
    inet6 fd17:625c:f037:2:46f2:420c:c586:3630/64 scope global temporary dynamic
        valid_lft 86248sec preferred_lft 14248sec
    inet6 fd17:625c:f037:2:a00:27ff:fe53:6f3/64 scope global dynamic mngtmpaddr
        valid_lft 86248sec preferred_lft 14248sec
    inet6 fe80::a00:27ff:fe53:6f3/64 scope link
        valid_lft forever preferred_lft forever
ovs-system: <BROADCAST,MULTICAST> mtu 1500 qdisc noop state DOWN group default
qlen 1000
    link/ether b6:93:c9:53:58:3f brd ff:ff:ff:ff:ff:ff
```

No.	Time	Source	Destination	Protocol	Length	Info
7	0.256400808	10.0.2.15	10.0.2.15	TCP	76	55076 → 5672 [SYN] Seq=0 W
8	0.256416939	10.0.2.15	10.0.2.15	TCP	56	5672 → 55076 [RST, ACK] Se
9	0.261236567	10.0.2.15	10.0.2.15	TCP	76	55082 → 5672 [SYN] Seq=0 W
10	0.261245063	10.0.2.15	10.0.2.15	TCP	56	5672 → 55082 [RST, ACK] Se
11	0.305556603	10.0.2.15	10.0.2.15	TCP	76	55086 → 5672 [SYN] Seq=0 W
12	0.305569407	10.0.2.15	10.0.2.15	TCP	56	5672 → 55086 [RST, ACK] Se
13	0.308239105	10.0.2.15	10.0.2.15	TCP	76	55094 → 5672 [SYN] Seq=0 W
14	0.308246119	10.0.2.15	10.0.2.15	TCP	56	5672 → 55094 [RST, ACK] Se
15	1.179515735	10.0.2.15	10.0.2.15	MySQL	81	Request Query
16	1.180459771	10.0.2.15	10.0.2.15	MySQL	124	Response TABULAR Response
17	1.180474369	10.0.2.15	10.0.2.15	TCP	68	56392 → 3306 [ACK] Seq=14
18	1.183047984	10.0.2.15	10.0.2.15	MySQL	4409	Request Query
19	1.190115512	10.0.2.15	10.0.2.15	MySQL	5470	Response TABULAR Response
20	1.197204482	10.0.2.15	10.0.2.15	MySQL	81	Request Query
21	1.197957296	10.0.2.15	10.0.2.15	MySQL	79	Response OK
22	1.246802973	10.0.2.15	10.0.2.15	TCP	68	56392 → 3306 [ACK] Seq=436

Frame 1: 76 bytes on wire (608 bits), 76 bytes captured on interface	0000	00 00 03 04 00 06 00 00	00 00 00 00 00 00 00 00
Ethernet II, Src: Linux cooked capture v1 (00:00:00:00:00:00), Dst: 00:00:00:00:00:00	0010	45 00 00 3c c8 52 40 00	40 06 5a 4c 0a 00 02 00
Internet Protocol Version 4, Src: 10.0.2.15, Dst: 10.0.2.15	0020	0a 00 02 0f d7 04 16 28	ad 37 60 8b 00 00 00 00
Transmission Control Protocol, Src Port: 55044, Dst Port: 5672	0030	a0 02 ff d7 18 4c 00 00	02 04 ff d7 04 02 00 00
	0040	e2 6e 4f 89 00 00 00 00	01 03 03 07

Result

IP spoofing was successfully demonstrated, showing how packet headers can be manipulated and emphasizing the need for strong network security measures.